① $f(1,1)=1$

$$\frac{\partial f}{\partial x}(x,y)=\frac{1}{x}+e^{x-2y+1} \Rightarrow \frac{\partial f}{\partial x}(1,1)=2$$

$$\frac{\partial f}{\partial y}(x,y)=\frac{1}{y}-2e^{x-2y+1} \Rightarrow \frac{\partial f}{\partial y}(1,1)=-1$$

$$\frac{\partial^2 f}{\partial x^2}(x,y)=-\frac{1}{x^2}+e^{x-2y+1} \Rightarrow \frac{\partial^2 f}{\partial x^2}(1,1)=0$$

$$\frac{\partial^2 f}{\partial y\partial x}(x,y)=-2e^{x-2y+1} \Rightarrow \frac{\partial^2 f}{\partial y\partial x}(1,1)=-2$$

$$\frac{\partial^2 f}{\partial y^2}(x,y)=-\frac{1}{y^2}+4e^{x-2y+1} \Rightarrow \frac{\partial^2 f}{\partial y^2}(1,1)=3$$

Tedy

$$T_2(x,y)=1+2(x-1)-(y-1)-2(x-1)(y-1)+\frac{3}{2}(y-1)^2$$

Odtud

$$f\left(\frac{9}{10},\frac{11}{10}\right)\approx T_2\left(\frac{9}{10},\frac{11}{10}\right)=1-\frac{2}{10}-\frac{1}{10}+\frac{2}{100}+\frac{3}{200}$$

$$=\frac{147}{200}$$

② • Podezřelé body z extrému v int M:

$$\frac{\partial f}{\partial x} = 4x - 4 = 0 \iff x = 1$$
$$\frac{\partial f}{\partial y} = 6y = 0 \iff y = 0$$

Zřejmě $(1,0) \in \operatorname{int} M$.

• Podezřelé body z extrému v ∂M:

$$4x - 4 + \lambda 2x = 0 \quad \ldots \quad (2+\lambda)x - 2 = 0 \quad (1)$$
$$6y + \lambda 2y = 0 \quad \ldots \quad (3+\lambda)y = 0 \quad (2)$$
$$x^2 + y^2 = 16 \quad \ldots \quad x^2 + y^2 = 16 \quad (3)$$

$(2) \iff \lambda = -3$ nebo $y = 0$

i) Je-li $\lambda = -3$, pak z (1) máme $x = -2$. Díky (3) je $y = \pm\sqrt{12}$.

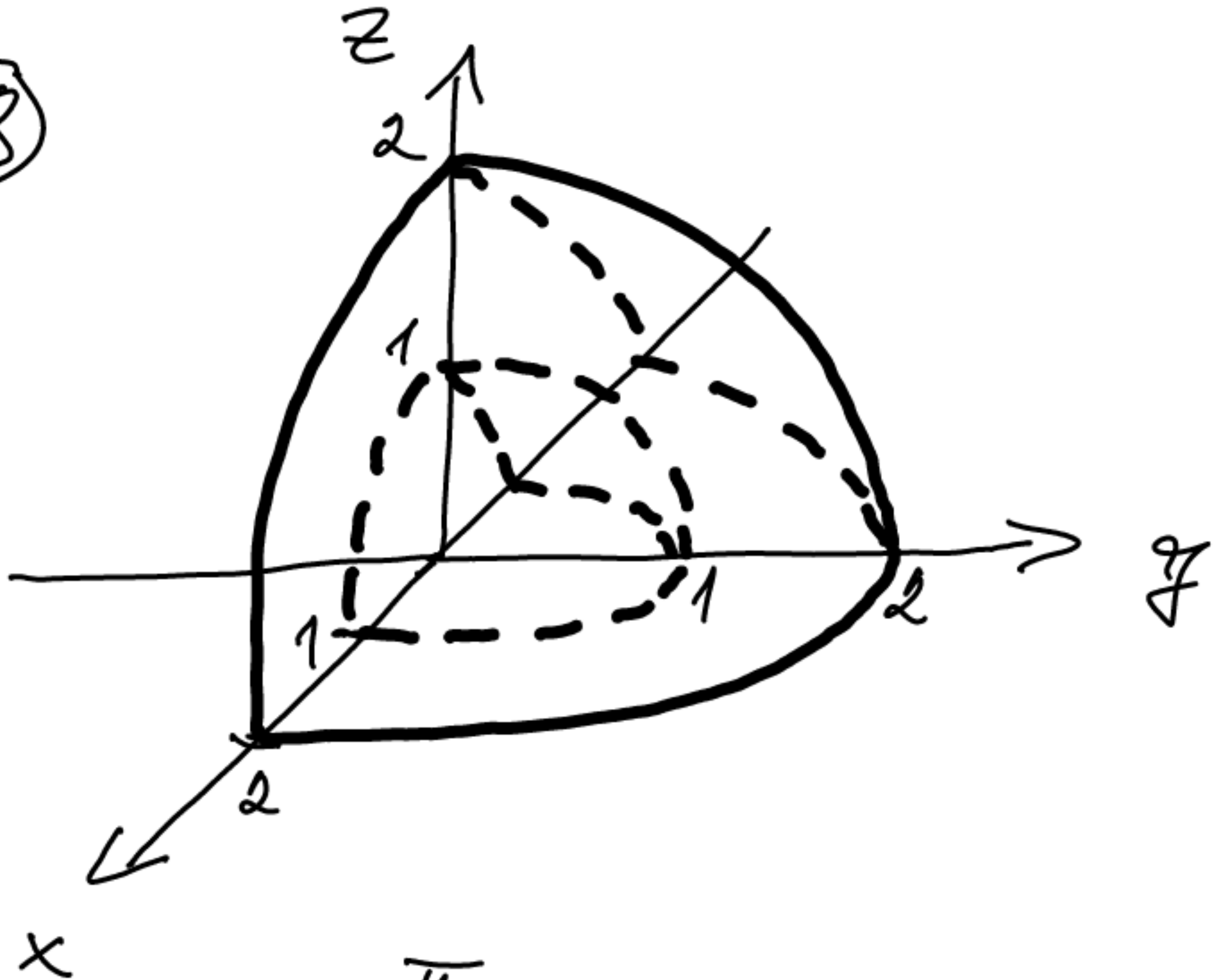
ii) Je-li $y = 0$, pak z (3) je $x = \pm 4$.

Podezřelé body v ∂M jsou $(-2, \pm\sqrt{12})$ a $(\pm 4, 0)$.

$$f(1,0) = -2$$
$$f(-2, \pm\sqrt{12}) = 8 + 36 + 8 = 52$$
$$f(4,0) = 32 - 16 = 16$$
$$f(-4,0) = 32 + 16 = 48$$

$\Rightarrow (-2, \sqrt{12}), (-2, -\sqrt{12})$ jsou body maxima a $(1,0)$ je bod minima f na M.

③



$$\int_M y = \int_0^{\frac{\pi}{2}} \int_0^{\pi} \int_1^{2} r \sin\varphi \sin\theta \, r^2 \sin\theta \, dr \, d\varphi \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{\pi} \int_1^{2} r^3 \sin\varphi \sin^2\theta \, dr \, d\varphi \, d\theta$$

$$= \left[\frac{r^4}{4}\right]_1^2 \left[-\cos\varphi\right]_0^{\pi} \int_0^{\frac{\pi}{2}} \sin^2\theta \, d\theta$$

$$= \left(4 - \frac{1}{4}\right) 2 \int_0^{\frac{\pi}{2}} \frac{1 - \cos(2\theta)}{2} \, d\theta$$

$$= \frac{15}{2} \left(\frac{\pi}{4} - \left[\frac{\sin(2\theta)}{4} \right]_0^{\frac{\pi}{2}} \right) = \frac{15\pi}{8}.$$

④ Ať $F(x,y) = (0,x)$. Pak

$$\text{obsah}(M) = \int_M 1 = \int_M \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \overset{\text{Greenova věta}}{=} \int_{\partial M} F$$

$$= \int_0^1 (0, t^2 - t) \cdot (2t-1, 3t^2 - 1)\, dt$$

$$= \int_0^1 3t^4 - t^2 - 3t^3 + t \, dt$$

$$= \frac{3}{5} - \frac{1}{3} - \frac{3}{4} + \frac{1}{2}$$

$$= \frac{36 - 20 - 45 + 30}{60} = \frac{1}{60}.$$

(5) $f(x) = \dfrac{2x+2}{(x-1)(x+3)} = \dfrac{1}{x-1} + \dfrac{1}{x+3}$

$$\frac{1}{x-1} = \frac{1}{(x+1)-2} = -\frac{1}{2}\,\frac{1}{1-\frac{x+1}{2}} = -\frac{1}{2}\sum_{k=0}^{\infty}\frac{(x+1)^k}{2^k}$$

pro $|x+1|<2$.

$$\frac{1}{x+3} = \frac{1}{(x+1)+2} = \frac{1}{2}\,\frac{1}{1-\left[-\frac{x+1}{2}\right]} = \frac{1}{2}\sum_{k=0}^{\infty}\frac{(-1)^k}{2^k}(x+1)^k$$

pro $|x+1|<2$.

Proto

$$f(x) = -\frac{1}{2}\sum_{k=0}^{\infty}\frac{(x+1)^k}{2^k} + \frac{1}{2}\sum_{k=0}^{\infty}\frac{(-1)^k}{2^k}(x+1)^k$$

$$= \sum_{k=0}^{\infty}\frac{(-1)^k-1}{2^{k+1}}(x+1)^k \quad \text{pro } |x+1|<2$$

(tj. $R=2$).